

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE**Paper 1 (Generated practice paper - Set 04)**

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours**Materials**

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
TOTAL	

Answer all questions in the spaces provided.

0 1

A student determined the water and organic matter content of a soil sample. Table 1 shows the results.

Table 1

Measurement	Mass / g
Empty crucible	25.0
Crucible + fresh soil	65.0
Crucible + soil after drying at 105 °C to constant mass	55.0
Crucible + soil after heating in a furnace at 550 °C	52.6

0 1

1

Calculate the percentage water content of the fresh soil.

[2 marks]

Answer

%

0 1

2

Calculate the percentage organic matter content of the dry soil.

[2 marks]

Answer

%

0 1

3

Explain why the soil is dried to constant mass before the organic matter is burnt off.

[2 marks]

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0 1 . 4

Explain how the organic matter content affects the fertility and erosion risk of this soil.

[3 marks]

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0 1 . 5

Suggest **one** reason why two students measuring the same field might get different organic matter results.

[1 mark]

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10

0 2

A student sampled the ground-flora of two woodlands. Table 1 shows the results for woodland A.

Table 1

Species	Number of individuals (n)
Bluebell	40
Wood anemone	30
Dog's mercury	20
Bramble	10
Total (N)	100

0 2 . 1

Use the formula $D = N(N - 1) / \sum n(n - 1)$ to calculate Simpson's index of diversity for woodland A.

[3 marks]

Answer

0 2 . 2

Woodland B had an index of 1.6. State what this shows and give **one** limitation of using the index alone to compare the woodlands.

[2 marks]

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5

0 3 . 1

Explain the difference between a negative and a positive feedback mechanism.

[2 marks]

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0 3 . 2

Describe **two** negative feedback mechanisms that may reduce global warming.

[4 marks]

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03 . 3

Explain how melting permafrost and melting methane hydrate act as positive feedback mechanisms.

[4 marks]

03 . 4

Give **three** other positive feedback mechanisms.

[3 marks]

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3

03 . 5

Explain what is meant by a climate **tipping point**.

[2 marks]

0 4 Table 1 shows the efficiency and cost of four types of solar photovoltaic (PV) cell.

Table 1

PV cell type	Efficiency / %	Relative cost per m ²
Amorphous silicon (thin film)	8	1
Polycrystalline silicon	16	2
Monocrystalline silicon with anti-reflective surface	22	3
Multi-junction cell	40	25

0 4 . **1** Describe how a photovoltaic cell generates electricity.

[2 marks]

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0 4 . **2** Explain how anti-reflective surfaces and multi-junction cells increase efficiency.

[3 marks]

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0 4 . **3** Use Table 1 to suggest which cell type would be chosen for a large solar farm on cheap desert land and which for a satellite. Justify your choices.

[3 marks]

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0 4 . 4

Explain why solar PV output is intermittent **and** unpredictable.**[2 marks]**

10

0 5 . 1

Which of the following statements about stratospheric ozone is correct? Tick one box.

[1 mark]

A Ozone is destroyed only by CFCs

☐

B Ozone absorbs mainly infrared radiation

☐

C Ozone is formed when monatomic oxygen reacts with diatomic oxygen

☐

D Ozone concentrations are highest at ground level

☐

0 5 . 2

Explain the difference between the roles of ozone in the stratosphere and in the troposphere.

[2 marks]

0 5 . 3

Explain why HCFCs cause less ozone depletion than CFCs.

[2 marks]

5

0 6

Two insecticides, X and Y, were tested on a pest insect and on a predatory ground beetle that feeds on the pest. Table 1 shows the results.

Table 1

Insecticide	LD ₅₀ for pest / mg kg ⁻¹	LD ₅₀ for ground beetle / mg kg ⁻¹	Half-life in soil / days	Solubility in water
X	5	6	180	Low
Y	8	150	4	High

0 6**. 1**

Explain what is meant by LD₅₀.

[1 mark]**0 6****. 2**

Using Table 1, explain which insecticide is more likely to cause indirect effects on the pest population through the food chain.

[3 marks]**0 6****. 3**

Suggest **two** environmental problems that could be caused by insecticide Y that are less likely with insecticide X.

[2 marks]**1****2****0 6****. 4**

Explain why the half-life of insecticide X makes it more likely to bioaccumulate.

[2 marks]

06 . 5

Give **two** reasons why a systemic insecticide may be preferred to a contact insecticide.

[2 marks]

1

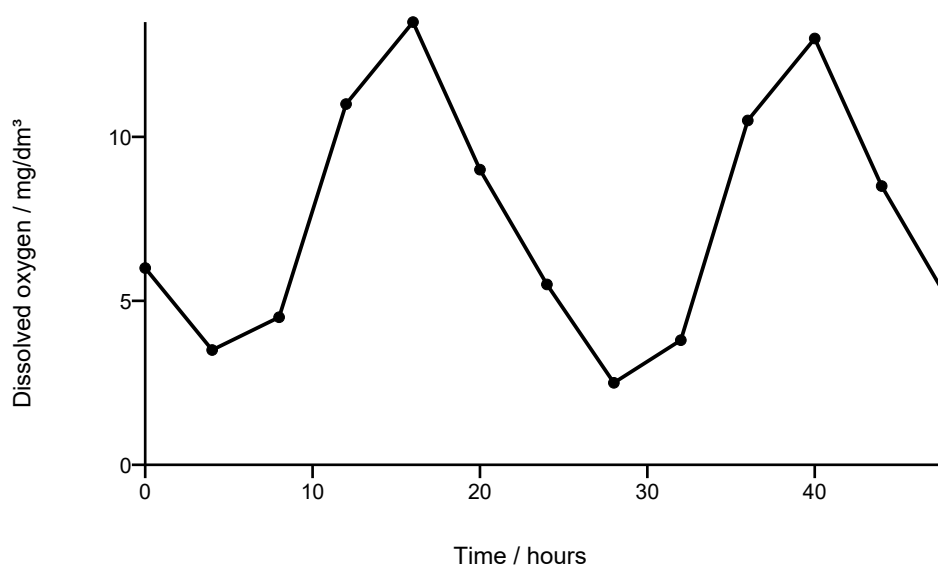
2

10

07

Figure 1 shows the dissolved oxygen concentration in a shallow lowland lake over 48 hours in late summer. The lake receives runoff from arable farmland and effluent from a sewage works.

Figure 1



07 . 1

Describe the pattern shown in Figure 1.

[2 marks]

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07 . 2

Explain the causes of the pattern shown in Figure 1.

[4 marks]

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07 . 3

Explain why fish kills in eutrophic lakes usually happen at night or at the end of summer.

[2 marks]

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07 . 4

Give **two** other harmful effects of algal blooms.

[2 marks]

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2

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10

08 . 1

Which of the following oil spill treatments is most likely to kill organisms living deep in beach sediments that were not affected by the oil? Tick one box.

[1 mark]

A Absorbent mops

☐

B Steam washing

☐

C Inflatable booms

☐

D Polymerising agents

☐

08 . 2

Explain how polymerising agents make an oil spill easier to clean up.

[2 marks]

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08 . 3

Explain why deoxygenation is a more common problem in water than in air.

[2 marks]

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5

09

Table 1 shows the concentration of cadmium in rice and in the kidneys of people living in two villages in Japan. Village A is downstream of a zinc mine.

Table 1

Village	Cadmium in rice / mg kg ⁻¹	Cadmium in kidney of residents / mg kg ⁻¹	Cases of bone disease (itai-itai) per 1000 people
A (downstream of mine)	1.2	180	34
B (control)	0.05	22	0

09 . 1

Calculate how many times more concentrated cadmium is in the kidneys of village A residents than in their rice.

[1 mark]

Answer times

0 9 . 2 Explain why cadmium and zinc pollution often occur together. **[1 mark]**

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0 9 . 3 Use Table 1 and your knowledge of the properties of cadmium to explain the high rate of bone disease in village A. **[4 marks]**

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0 9 . 4 Suggest **two** ways cadmium pollution from the mine could be controlled. **[2 marks]**

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0 9 . 5 Cadmium is used in cadmium-telluride photovoltaic panels. Suggest why this new use may cause pollution in the future. **[2 marks]**

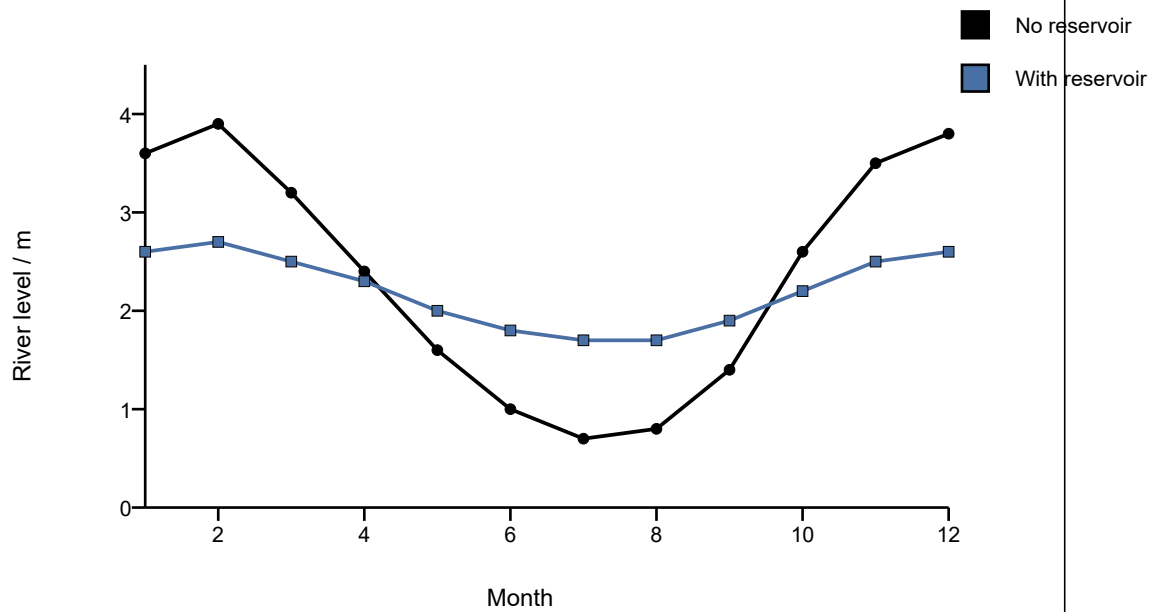
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10

1 0

Figure 1 shows the river level below a reservoir site during one year, before and after a river-regulation reservoir was built.

Figure 1



1 0

1

Use Figure 1 to calculate the reduction in the annual range of river level produced by the reservoir.

[2 marks]

Answer

m

1 0

2

Explain how the reservoir produces the changes shown in Figure 1.

[2 marks]

1 0 . 3

Explain why human activities in the catchment may have increased the extremes in river flow before the reservoir was built.

[2 marks]

1 0 . 4

Discuss the factors that should be considered when selecting a site for a new reservoir, and evaluate the alternative strategies for increasing the sustainability of water supplies.

[9 marks]

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Extra space

15

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Write an essay on one of the following topics.

1	1	.	1
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Discuss the extent to which energy conservation technologies could reduce the need for new energy supplies in a developed country.

[25 marks]

OR

1 1 . 2

Discuss how knowledge of the properties of energy resources can be used to reduce the severity of their impact on the environment.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

[illegible]

[illegible]

END OF QUESTIONS